

Assignment 1

Explore following 4 websites and answer the questions

1. bitcoin.org
 - a. Official educational website for learning about Bitcoin and how it works.
 - b. Explains Bitcoin wallets, mining, blockchain, and peer-to-peer transactions.
 - c. Provides beginner guides, Bitcoin whitepaper, and security tips.
 - d. Helps users choose wallets and learn how to buy or sell Bitcoin safely.
 - e. Open-source, community-funded project that promotes Bitcoin adoption and awareness.
2. [Blockchain.com](https://blockchain.com)
 - a. Provides a crypto wallet to store, send, and receive cryptocurrencies.
 - b. Offers a Blockchain Explorer to track Bitcoin and other blockchain transactions.
 - c. Supports buying, selling, and trading cryptocurrencies.
 - d. Serves both individual users and institutional investors with crypto infrastructure.
 - e. One of the oldest blockchain companies, founded in 2011, with millions of wallets created.
3. <https://coinmarketcap.com/>
 - a. Tracks live prices of thousands of cryptocurrencies.
 - b. Displays market capitalization, trading volume, and ranking of crypto coins.
 - c. Provides historical price charts and market analysis.
 - d. Lists cryptocurrency exchanges and trading pairs.
 - e. Offers crypto news, educational content, portfolio tracking, and watchlists.
4. <https://www.investopedia.com/>
 - a. One of the most trusted websites for learning finance and investing concepts.
 - b. Explains financial terms through an extensive dictionary and tutorials.
 - c. Covers topics like stocks, cryptocurrencies, mutual funds, banking, and personal finance.
 - d. Publishes market news, investment strategies, and educational articles.
 - e. Content is reviewed by financial experts to ensure accuracy and reliability.

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom?
Answer: Satoshi Nakamoto.
2. Bitcoin transactions are recorded on which data structure?
Answer: Blockchain.
3. Bitcoin uses which consensus mechanism?
Answer: Proof-of-Work (PoW).
4. The smallest unit of Bitcoin is called?
Answer: Satoshi.
5. Bitcoin is a **peer-to-peer** type of what system?
Answer: Payment System

Wallets

6. A wallet stores the user's _____ key.
Answer: Private.
7. The recovery phrase used in wallets is called a _____ phrase.
Answer: Seed.
8. Hardware wallets store keys _____ (online/offline).
Answer: Offline.
9. Bitcoin addresses are derived from _____ keys.
Answer: Public.
10. What is required to sign a transaction? (public/private)
Answer: Private.

Security

11. Bitcoin prevents double-spending using _____.
Answer: Blockchain.

12. Bitcoin is secured using _____ cryptography (asymmetric/symmetric).

Answer: Asymmetric.

13. To run a Bitcoin node, the software required is Bitcoin _____.

Answer: Core.

14. The process of verifying transactions is called _____.

Answer: Validation.

15. The Bitcoin network is described as _____ (centralized/decentralized).

Answer: Decentralized.

Mining

16. Miners solve _____ puzzles to add blocks.

Answer: Cryptographic.

17. Mining rewards a block with newly created _____.

Answer: Bitcoins.

18. The block time target for Bitcoin is _____ minutes.

Answer: 10.

19. Mining difficulty adjusts every _____ blocks.

Answer: 2016.

20. Miners compete to find a valid block _____ (hash/signature).

Answer: Hash.

Blockchain Structure

21. Each block contains a _____ root.

Answer: Merkle.

22. Blocks are linked using the previous block's _____.

Answer: Hash.

23. A Bitcoin transaction contains inputs and _____.

Answer: Outputs.

24. Unspent transaction outputs are called _____.

Answer: UTXOs.

11. Number of inputs: **1**

12. Number of outputs: **2**

13. Transaction fee: **0.00004512 BTC**

14. How many confirmations does it have?

Answer : 105,420+

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block.

16. Record:

- Total BTC received: **1,250.45 BTC**
- Total BTC sent: **1,200.00 BTC**
- Current balance: **50.45 BTC**

What is the role of a full node?

Answer : A full node downloads and independently verifies the *entire* Bitcoin blockchain against the protocol's consensus rules — checking every block and transaction for validity (correct signatures, no double-spends, valid proof-of-work, etc.). It relays valid transactions and blocks to other nodes on the network, maintains a complete, trustless copy of the ledger, and enforces the rules of the network — if a miner or another node tries to break a rule (e.g., create coins out of thin air or approve an invalid transaction), a full node will reject it. This is what gives Bitcoin its decentralization: no single party needs to be trusted, because anyone running a full node can verify the truth of the chain for themselves.

List any two developer resources available on bitcoin.org.

Answer :

1) Bitcoin Developer Guide — a conceptual walkthrough of how Bitcoin works under the hood (transactions, blocks, the P2P network, wallets, mining, etc.), aimed at people building Bitcoin software.

2) Bitcoin Developer Reference — detailed technical reference material, including the Bitcoin Core RPC API, P2P network message formats, script opcodes, and other protocol specifications.

Explore : <https://coinmarketcap.com/>

Task C1: Market Information

- Current price of BTC: **\$64,535.93**.
- 24-hour % change: **+0.24%**.
- Market capitalization: **\$1.29 Trillion**.
- Circulating supply: **20.06 Million BTC**.
- Total/maximum supply: **20.06 Million / 21 Million BTC**.
- **Task C2: Historical Data**
- What was the price of Bitcoin one year ago?
Answer: Approximately **\$64,000–\$65,000**.
- Note the highest and lowest prices in the last 30 days.
Answer : High: ~\$70,200 USD | Low: ~\$62,100 USD

Part D – Investopedia

Explore <https://www.investopedia.com/>

1) Why is Bitcoin considered volatile?

Answer: Because its price changes rapidly due to supply and demand, investor sentiment, regulations, and market news.

2) List two investment risks mentioned.

Answer:

1. Market volatility.
2. Regulatory uncertainty.

Part E: Reflection & Analysis

1) Which website gave you the most clear understanding? Why?

Answer : Bitcoin.org, because it explains Bitcoin concepts, wallets, mining, security, and blockchain in simple language suitable for beginners.

2) In one paragraph, explain how blockchain ensures trust without a central authority.

Answer : Blockchain ensures trust without a central authority by storing transactions on a distributed ledger shared across thousands of computers (nodes). Every transaction is verified through a consensus mechanism before being added to the blockchain. Once recorded, the data cannot be changed without the agreement of the majority of the network, making it secure and tamper-resistant. This decentralized verification process removes the need for banks or intermediaries while maintaining transparency and trust.